PAISAPATH AI: Personal AI Financial Co-Pilot

Project Documentation:

PaisaPath AI is a personal finance management web application designed to help users organize, monitor, and understand their financial activities through a single platform. The application allows users to record their income and expenses, manage budgets, create financial goals, monitor progress, and view financial reports through graphical representations. It also includes an AI Financial Advisor that provides explanations and guidance based on the user's financial information.

The main objective of PaisaPath AI is to simplify personal financial management by bringing important financial information together in one place. Instead of maintaining separate records for income, expenses, savings, and goals, users can manage these activities through an integrated interface.

A major design principle of the application is the use of a centralized Financial Calculation Engine. Financial values such as total income, total expenses, savings, savings rate, budget comparison, and goal progress are calculated through this centralized system. The user interface displays the verified results, while the AI Advisor uses the verified financial information to provide explanations.

1. Scenarios

Scenario 1: The first scenario demonstrates how a user can maintain their regular financial records using PaisaPath AI. The user can enter different sources of income and record expenses according to categories such as rent, food, education, transport, shopping, healthcare, utilities, insurance, travel, and personal expenses.

The user begins by entering their income information, including the amount, source, frequency, recurring status, date, and description. The same approach is used for expenses, where the user can select an appropriate category and payment method. Once the records are saved, the application processes the financial information and uses it to generate financial summaries and reports.

For example, a user may have a monthly income of ₹1,00,000 and monthly expenses of ₹55,000. The system can determine the resulting savings of ₹45,000 and calculate the corresponding savings rate. These verified values can then be displayed in the financial reports and used by other modules of the application.

Activities

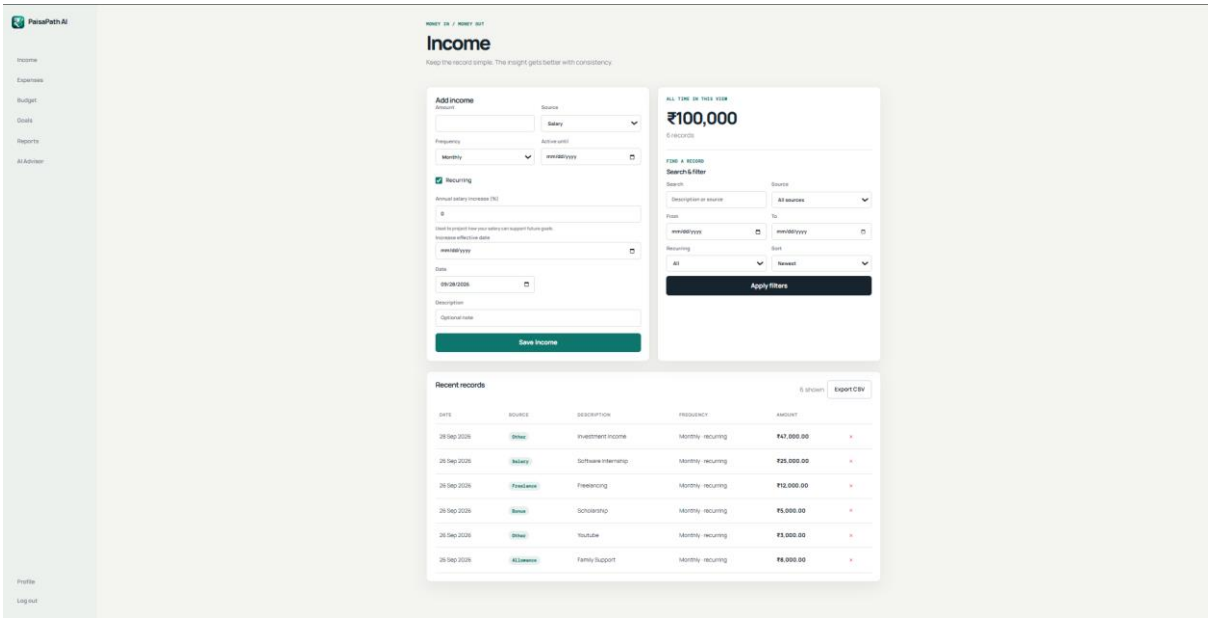
Activity 1.1: The user enters income records with the required financial details.

Activity 1.2: The user records expenses and assigns appropriate categories.

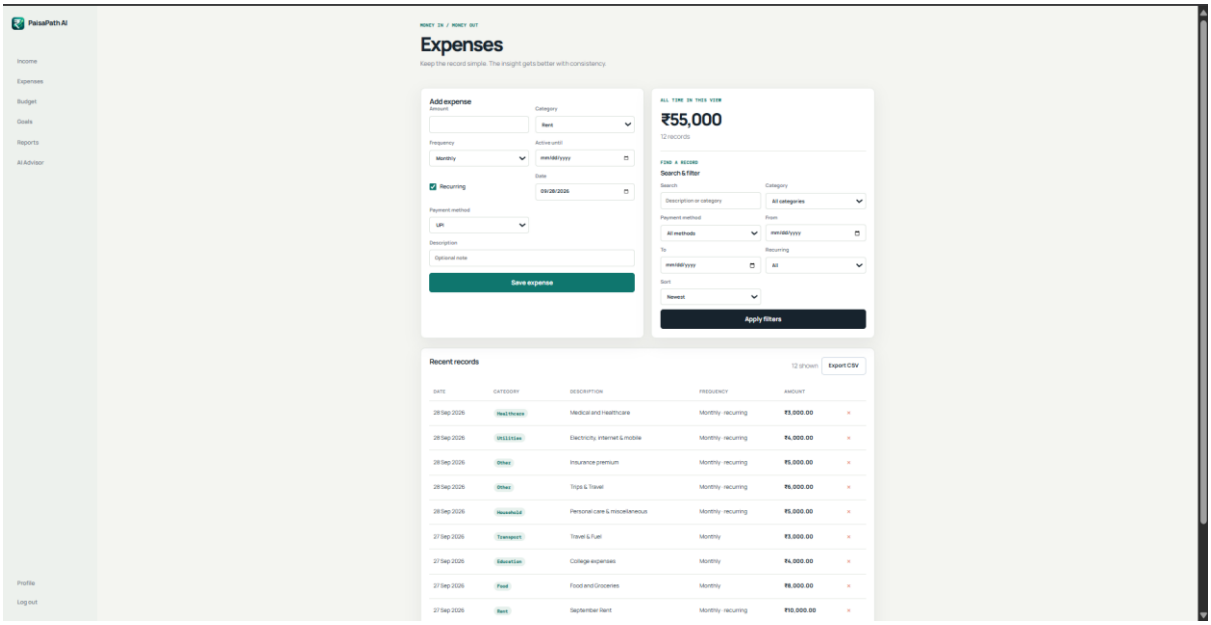
Activity 1.3: The application validates and processes the financial records.

Activity 1.4: The Financial Calculation Engine calculates the user's financial summary.

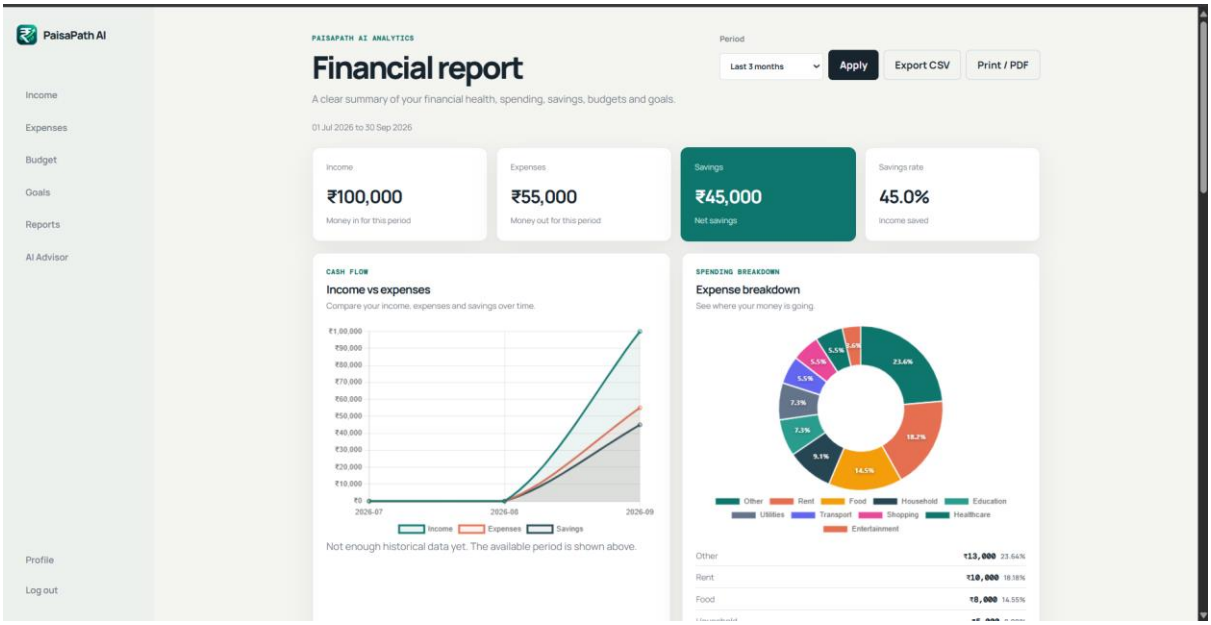
Activity 1.5: The resulting information is displayed through financial reports and visualizations.



Description: The Income page provides a structured form for entering income records and displays previously added income records. Users can specify the income source, amount, frequency, recurring status, date, and description.



Description: The Expenses page allows users to record their spending by selecting an expense category, entering the amount, choosing the frequency and payment method, and adding an optional description.



Description: The Financial Report presents processed financial information through summary values and graphical representations, allowing users to understand the relationship between income, expenses, and savings.

Scenario 2: The second scenario demonstrates how users can create and monitor financial goals. PaisaPath AI allows users to define a target amount, specify how much has already been saved, select a target date, and provide additional notes about the purpose of the goal.

A goal can represent different financial requirements such as an emergency fund, laptop purchase, education expenses, travel, investments, or a future home down payment. After the goal is created, the application processes the available information to determine the remaining amount and progress toward the target.

The goal tracking system helps users understand how much of the target has already been achieved and how much remains to be saved. The target date is also used to provide information about the required savings needed to work toward the goal.

Activities

Activity 2.1: The user opens the Goals section and creates a new financial goal.

Activity 2.2: The user enters the target amount and amount already saved.

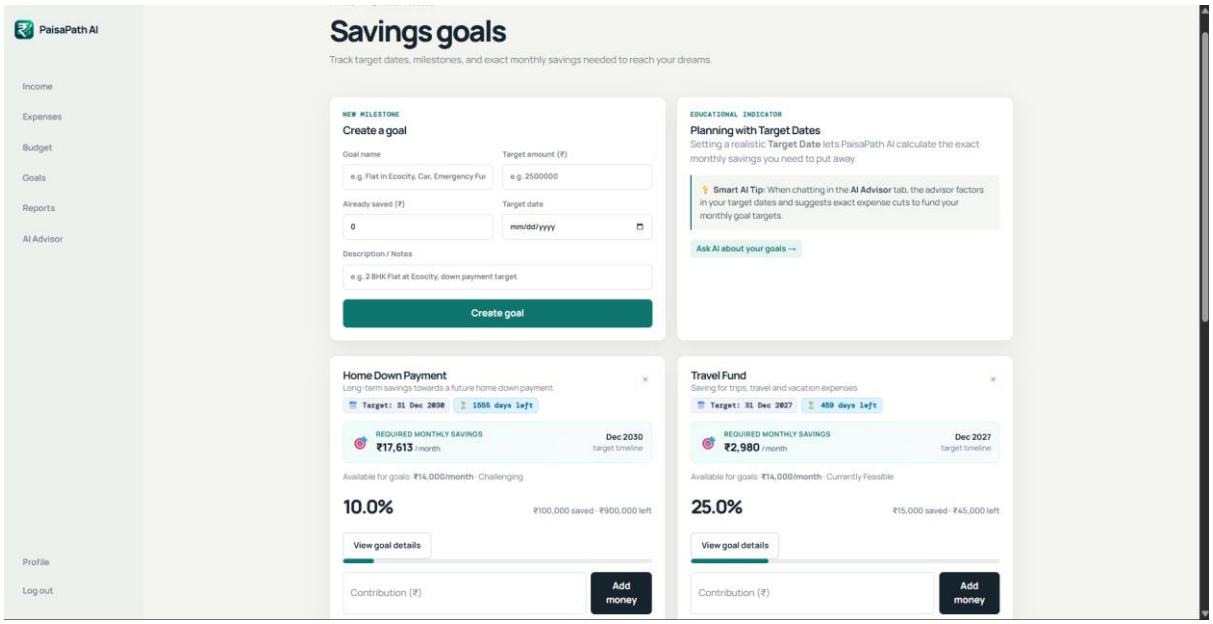
Activity 2.3: The user enters the target date and goal description.

Activity 2.4: The application calculates the remaining amount and goal progress.

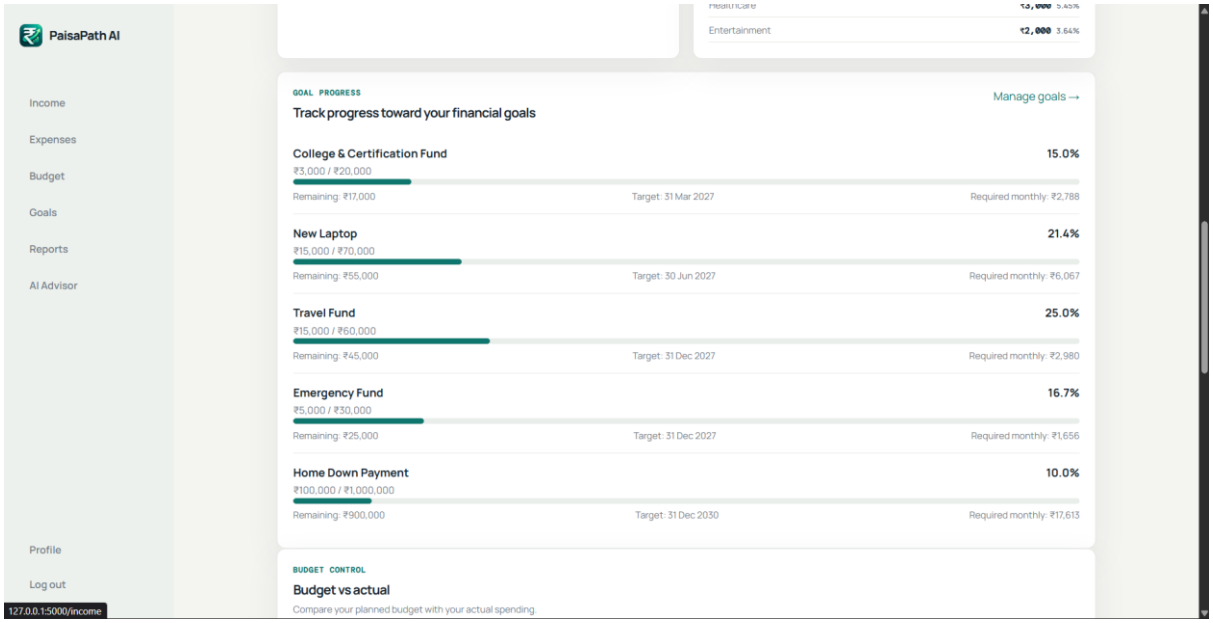
Activity 2.5: The calculated goal information is displayed through the goal interface.

Example

The user has created multiple financial goals, including a **Home Down Payment** goal with a target amount of **₹10,00,000** and **₹1,00,000 already saved**. The application shows **₹9,00,000 remaining**, with a target timeline up to **December 2030** and a required monthly saving of approximately **₹17,613**. Based on the available amount for goals of **₹14,000 per month**, the goal is marked as **Challenging**. The user has also created a **Travel Fund** goal with a target amount of **₹60,000**, of which **₹15,000 has already been saved**. The remaining amount is **₹45,000**, with a target of **December 2027** and required monthly savings of approximately **₹2,980**. Since the available amount for goals is ₹14,000 per month, the Travel Fund is currently shown as **Feasible**.



Description: The Goals page provides a goal creation form where users can enter the goal name, target amount, existing savings, target date, and description. Existing goals are displayed with their progress and target information.



Description: The goal section provides users with a visual representation of their progress and information related to the target amount and remaining savings requirement.

Scenario 3: The third scenario demonstrates the AI Financial Advisor of PaisaPath AI. The AI Advisor is designed to help users understand their financial information in a conversational manner.

The financial information is first processed by the application's Financial Calculation Engine. Verified financial results are then made available to the AI Advisor so that it can explain the user's financial situation in a more understandable way.

For example, if the application calculates that a user has ₹1,00,000 in monthly income, ₹55,000 in expenses, and ₹45,000 in savings, the AI Advisor can use these verified values to explain the user's financial position and provide relevant financial guidance.

The AI Advisor is therefore used primarily for explanation and assistance, while the Financial Calculation Engine remains responsible for the actual financial calculations.

Activities

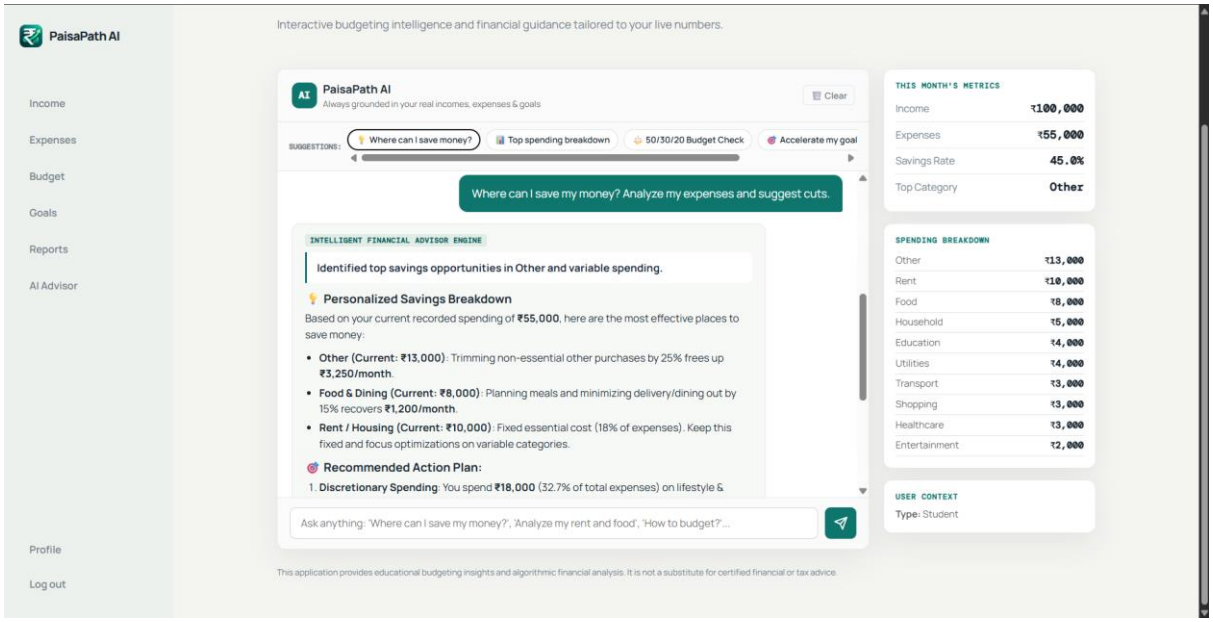
Activity 3.1: The user records financial information in the application.

Activity 3.2: The Financial Calculation Engine processes the information.

Activity 3.3: Verified financial results are generated.

Activity 3.4: The user asks a financial question through the AI Advisor.

Activity 3.5: The AI Advisor uses the verified financial context to provide an explanation.



Description: The AI Advisor interface provides a conversational environment where users can ask questions related to their financial information. The advisor uses the available verified financial context to explain financial information and provide guidance.

2. Technical Architecture

2.1 Architecture of PaisaPath AI

PaisaPath AI follows a modular web application architecture designed to manage personal financial information and provide AI-assisted financial guidance. The architecture connects the user interface, authentication/session management, financial data, centralized calculation engine, reports and charts, and AI Financial Advisor. The system is designed so that financial calculations are performed centrally and the resulting verified values are used consistently throughout the application.

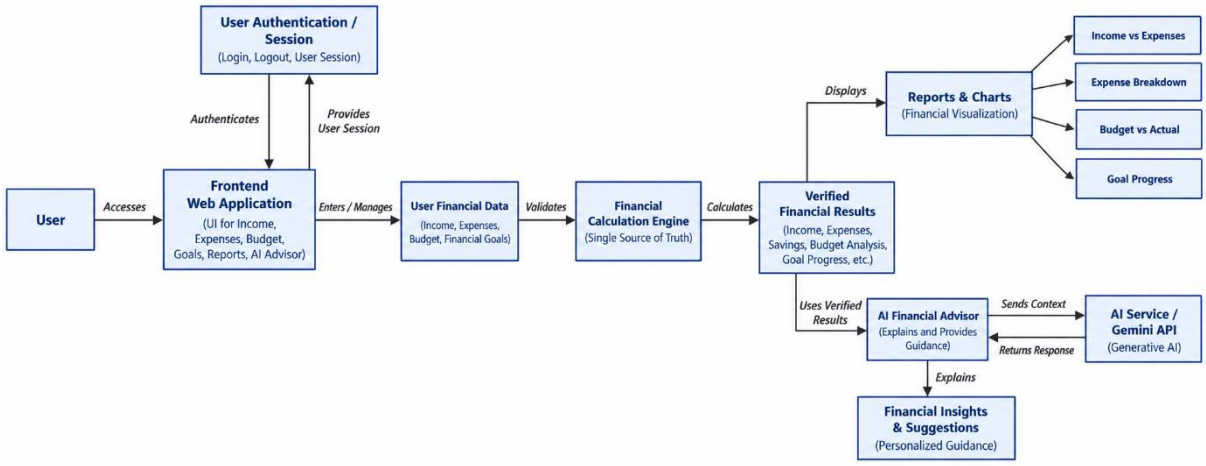


Figure 1: Technical Architecture of PaisaPath AI

The user interacts with PaisaPath AI through the frontend web application. The frontend provides interfaces for managing **Income, Expenses, Budget, Goals, Reports, and AI Advisor**. The user enters or manages financial information through these interfaces, while the application handles the processing and presentation of the resulting information.

2.2 User Authentication and Session

The authentication and session layer manages the user's active application session. After authentication, the frontend receives the relevant user session and uses it while accessing the application's financial features. This allows financial information to be associated with the appropriate user and helps prevent financial records from different users from being mixed.

2.3 Frontend Web Application

The frontend is the main interaction layer of PaisaPath AI. It provides the user interface for the application's financial modules. Users can enter income records, categorize expenses, create budgets, establish financial goals, view reports, and interact with the AI Financial Advisor.

The frontend also displays calculated financial information through cards, tables, progress indicators, and charts. It presents verified results generated by the application's financial processing system rather than independently creating conflicting financial calculations.

2.4 User Financial Data

The application processes different types of financial information entered by the user. This includes income records, expense records, budget information, and financial goals.

Income information can include the amount, source, frequency, recurring status, date, and description. Expense information can include the amount, category, frequency, payment method, date, recurring status, and description. Financial goals contain information such as the goal name, target amount, amount already saved, target date, and description.

2.5 Financial Calculation Engine

The **Financial Calculation Engine** is the central component of the architecture and acts as the **single source of truth** for financial calculations.

Validated financial information is passed to the calculation engine, where the required financial metrics are calculated. These can include total income, total expenses, net savings, savings rate, budget comparisons, remaining goal amounts, goal progress, and required savings for financial goals.

The centralized approach ensures that the same financial values are used across the Reports, Goals, and AI Advisor sections. This reduces inconsistencies that could occur if individual frontend components performed their own calculations.

2.6 Verified Financial Results

After processing the user's financial information, the Financial Calculation Engine produces verified financial results. These results form the common data source for the application's reporting and AI features.

For example, if the application calculates a monthly income of ₹1,00,000 and expenses of ₹55,000, the verified net savings value is ₹45,000. This result can then be displayed in the Reports section and provided as financial context to the AI Advisor.

2.7 Reports and Charts

The Reports and Charts component presents verified financial results visually. The Reports section helps users understand their financial position through financial summaries and graphical representations.

The application can display information such as:

- **Income vs Expenses**
- **Expense Breakdown**
- **Budget vs Actual**
- **Goal Progress**

These visualizations use the verified financial results generated by the Financial Calculation Engine.

2.8 AI Financial Advisor

The AI Financial Advisor provides a conversational interface through which users can ask questions about their financial situation. The advisor uses the user's relevant verified financial information as context when generating explanations and guidance.

The AI Advisor is not treated as the source of financial calculations. Numerical financial values are generated by the Financial Calculation Engine first, and the AI Advisor uses those results to explain the user's financial situation in a more understandable way.

2.9 AI Service / Gemini API

The AI Financial Advisor communicates with the configured generative AI service to generate natural-language responses. Relevant verified financial context is provided to the AI service, and the generated response is returned to the AI Advisor interface.

The AI service is therefore used primarily for **financial explanation and conversational guidance**, while the application's Financial Calculation Engine remains responsible for financial calculations.

2.11 Architectural Principle

The central architectural principle of PaisaPath AI is:

User Data → Validation → Financial Calculation Engine → Verified Results → UI + AI Explanation

3. Project Workflow

The development of PaisaPath AI was organized into six major milestones. Each milestone focused on a specific stage of application development, starting from requirement analysis and architecture and continuing through implementation, AI integration, testing, and documentation.

Milestone 1: Requirement Analysis, Model Selection and Architecture

The first milestone focused on understanding the requirements of the personal finance application and deciding how the major components should interact with each other. The project requirements included income management, expense tracking, budgeting, financial goals, financial reports, visualizations, and an AI Financial Advisor.

During this milestone, the application architecture was planned around a centralized financial calculation system. The relationship between the frontend, financial data, calculation engine, reporting modules, and AI Advisor was defined so that financial calculations would have a single reliable source.

Activities

Activity 1.1: Analyse the requirements of the personal finance management application.

Activity 1.2: Identify the major application modules such as Income, Expenses, Budget, Goals, Reports, and AI Advisor.

Activity 1.3: Design the application architecture and define the interaction between the frontend, backend/data layer, calculation engine, and AI integration.

Activity 1.4: Select and configure the required development libraries, frameworks, and AI integration components.

Milestone 2: Core Financial Functionality Development

The second milestone focused on implementing the core financial management functionality of PaisaPath AI. The income and expense modules were developed to allow users to create and manage financial records.

The application was designed to support different income sources and expense categories. Users can enter amounts, frequencies, recurring information, dates, descriptions, and other relevant financial details. The budget and goal modules were also developed to allow users to organize their financial plans.

Activities

Activity 2.1: Develop the Income module for adding and managing income records.

Activity 2.2: Develop the Expenses module for recording categorized expenses.

Activity 2.3: Implement budget management and budget comparison functionality.

Activity 2.4: Implement financial goal creation and progress tracking.

Milestone 3: Application and Data Processing Development

The third milestone focused on connecting the application modules with the underlying data processing system. Financial records entered through the interface needed to be stored and retrieved correctly so that they could be used throughout the application.

User-specific financial data handling was also considered during this stage. The system was structured so that income, expenses, budgets, goals, and related financial information belong to the appropriate user.

Activities

Activity 3.1: Connect application interfaces with financial data operations.

Activity 3.2: Implement validation and processing of financial records.

Activity 3.3: Implement user-specific financial data handling.

Activity 3.4: Connect financial data with the centralized calculation engine.

Milestone 4: Frontend Development and Visualization

The fourth milestone focused on developing the user interface of PaisaPath AI. The interface was designed to provide a clean and organized financial management experience.

Dedicated pages were created for income, expenses, budgets, goals, reports, and AI Advisor functionality. The Reports section was developed to present financial information through summary cards and charts.

The interface was also designed to provide meaningful empty states and organized financial information so that users can understand their records without unnecessary complexity.

Activities

Activity 4.1: Design and develop the user interface for the major application modules.

Activity 4.2: Develop forms for entering income, expenses, budgets, and goals.

Activity 4.3: Develop the Financial Reports interface.

Activity 4.4: Integrate charts and visualizations for financial information.

Activity 4.5: Improve responsiveness and usability of the application interface.

Milestone 5: AI Financial Advisor, Testing and Integration

The fifth milestone focused on integrating the AI Financial Advisor with the completed financial management system. The AI Advisor was designed to work with verified financial information generated by the application's calculation engine.

Testing was performed across the major financial workflows to verify that calculations, charts, reports, goals, and AI responses worked correctly with different financial datasets.

The application was also tested with multiple income sources, expense categories, budgets, and financial goals to ensure that the financial modules could handle different types of user data.

Activities

Activity 5.1: Integrate the AI Financial Advisor with the application.

Activity 5.2: Connect verified financial results with the AI Advisor context.

Activity 5.3: Test income and expense calculations.

Activity 5.4: Test budget comparisons and goal calculations.

Activity 5.5: Test financial reports and graphical visualizations.

Activity 5.6: Test user-specific financial data handling and overall application workflow.

Milestone 6: Documentation, Final Testing and Conclusion

The sixth milestone focused on final testing, documentation, and preparation of the completed project for submission.

The application was reviewed across its major modules, including income, expenses, budget, goals, reports, and AI Advisor. Different financial values and scenarios were used during testing to verify that the calculation engine and application interfaces produced consistent results.

Project screenshots were captured for important application pages and scenarios. Technical architecture, project workflow, scenarios, and final project information were documented for the project repository.

Activities

Activity 6.1: Perform final functional testing of the application.

Activity 6.2: Verify financial calculations and report values.

Activity 6.3: Capture screenshots of important application pages.

Activity 6.4: Prepare the three project scenarios.

Activity 6.5: Prepare technical architecture and six-milestone workflow documentation.

Activity 6.6: Organize the documentation folder and prepare the project repository for submission.

4. Website Pages / Modules

4.1 Income Page

The Income page is designed to help users maintain a clear record of their different sources of income. Users can enter important information such as the income amount, source, frequency, recurring status, date, and description. The page also displays previously added income records, allowing users to maintain a consistent history of their earnings. This information is processed by the application and used in financial calculations, reports, and other relevant modules.

4.2 Expenses Page

The Expenses page allows users to record and organize their spending according to different categories. Users can enter the expense amount, category, frequency, recurring status, payment method, date, and description. The application supports expense categories such as rent, food, education, transportation, shopping, entertainment, healthcare, utilities, insurance, travel, and personal expenses. These records are used to calculate total expenses and provide a clearer understanding of the user's spending patterns.

4.3 Budget Page

The Budget page helps users plan their spending by defining budget amounts for different financial categories. The application compares the planned budget with actual recorded expenses, allowing users to understand how their spending compares with their planned limits. This information is also used in the financial reports to provide a visual representation of budget performance and help users identify categories where actual expenses differ from planned amounts.

4.4 Goals Page

The Goals page allows users to create and track personal financial targets. A user can specify the goal name, target amount, amount already saved, target date, and description or notes. The application calculates the remaining amount and tracks the user's progress toward the target. It also uses the available financial information and target date to determine the savings requirements associated with the goal, helping users understand whether they are progressing toward their financial targets.

4.5 Reports Page

The Reports page provides a consolidated view of the user's financial information by presenting important financial metrics and visualizations. It displays information such as income, expenses, savings, and savings rate and provides graphical representations including Income vs Expenses, Expense Breakdown, Budget vs Actual, and Goal Progress. The information displayed in the Reports section is generated from the verified financial results produced by the Financial Calculation Engine.

4.6 AI Advisor Page

The AI Advisor provides a conversational interface through which users can ask questions related to their personal finances. The advisor uses the verified financial context generated by the application to explain the user's financial situation and provide relevant guidance. The Financial Calculation Engine remains responsible for calculating financial values, while the AI Advisor focuses on explaining those results and presenting them in an understandable form.

5. Conclusion

PaisaPath AI successfully combines personal finance management with AI-assisted financial guidance in a single web application. The application provides functionality for recording income and expenses, managing budgets, creating financial goals, viewing financial reports, and interacting with an AI Financial Advisor.

The project places particular emphasis on maintaining consistency in financial calculations through a centralized Financial Calculation Engine. This architecture ensures that financial values are calculated through a common source before being displayed in reports or explained by the AI Advisor.

The development process was completed through six milestones covering requirement analysis and architecture, core financial functionality, application and data processing, frontend development, AI integration and testing, and final documentation. The completed application was tested using multiple financial records and financial goal scenarios to verify the major workflows.

Overall, PaisaPath AI demonstrates how a modern financial management application can combine structured financial data, centralized calculations, visual reports, goal tracking, and AI-based explanations to provide users with a unified personal finance experience.